Name_____ Date_____ Period_____

LESSON 6-4 · HOMEWORK PACKET · SUPPORTED · SESSION 1 · PAGES 1–2

Writing Numerical Expressions

Maryland Standard 6.EE.A.2 · Learning target – I can write a numerical expression that matches a situation.

REMEMBER

- A **numerical expression** uses numbers and operations. It has no equals sign.
- Sum (+), difference (−), product (×), quotient (÷). **Less than** flips the order: 5 less than 12 is 12 − 5.
- Use **parentheses** for a part done first: the sum of 4 and 6, tripled, is 3 × (4 + 6).

WORKED EXAMPLE

Kayla buys 3 notebooks at $4 each and a backpack for $15. Write an expression for the total cost. Then evaluate it.

Step 1 Notebooks: 3 × 4. Backpack: 15

Step 2 Join them: 3 × 4 + 15

Step 3 Evaluate: 12 + 15 = 27

Answer **3 × 4 + 15 = $27**

WORD BANK

expression – numbers and operations with no equal sign

product – the answer when you multiply (×)

less than – subtract, and flip the order: 12 less than 50 is 50 − 12

parentheses ( ) – do this part first

PRACTICE

show your work · label units

1. Write a numerical expression for each phrase. Do not evaluate.

Step 1 Find the operation words: product, less than, quotient.

Step 2 Write the numbers in the right order.

a. the product of 7 and 8, minus 10 _____

b. 12 less than 50 _____

c. the quotient of 36 and 9, plus 2 _____

d. the difference of 15 and 6, doubled _____

e. 9 more than the product of 3 and 5 _____

f. the quotient of 42 and 7, tripled _____

2. **Field Day.** A class buys **5 packs** of water at **$6 each** and **4 bags** of oranges at **$3 each**. Write one expression for the **total cost**. Then evaluate it.

   Step 1 Water: 5 × 6. Oranges: 4 × 3.

   Step 2 Add the two products, then evaluate.

   Show your work.

   Expression and cost_____

3. **Bake Sale.** Nia sells 12 muffins at $2 each and 8 loaves of bread at $5 each. She spent $15 on supplies. Write one expression for the money she has left after paying for supplies. Then evaluate it.

   Step 1 Write a product for the muffins and one for the bread.

   Step 2 Add them, then subtract the $15 for supplies.

   Show your work.

   Expression and amount_____

4. Which expression matches "the sum of 8 and 2, times 5"?

   □ 8 + 2 × 5 □ (8 + 2) × 5 □ 8 × 5 + 2

5. **Write a Story.** Write a short story that matches 4 × (6 + 3). Then evaluate the expression.

   Frame ___ *friends each buy something for $6 and something for $3. The total is* ___.

CHECK YOURSELF	□ I can do this on my own	□ I need a little help	□ I have a question about #___

AT HOME Find two prices at home or in an ad. Write one expression for buying some of each.

Name_____ Date_____ Period_____

LESSON 6-4 · HOMEWORK PACKET · SUPPORTED · SESSION 2 · PAGES 3–4

Order of Operations with Powers

Maryland Standard 6.EE.A.2c · Learning target – I can evaluate an expression that contains grouping symbols and powers using the order of operations.

REMEMBER

- Order: **1.** grouping ( ) **2.** powers **3.** × and ÷ left to right **4.** + and − left to right.
- × and ÷ are partners: do whichever comes first from the left. So are + and −.
- An exponent applies to its base, including a group in parentheses: $2 \times 3^2 = 2 \times 9 = 18$.

WORKED EXAMPLE

Evaluate $20 - (2 + 1)^2 \times 2$.

Step 1 Grouping, then power: $20 - 3^2 \times 2 = 20 - 9 \times 2$

Step 2 Multiply, then subtract: $20 - 18 = 2$

Answer **2**

WORD BANK

order of operations – ( ) first, then powers, then × ÷, then + −

evaluate – find the value

power – a repeated factor, like $3^2 = 3 \times 3$

PRACTICE

show your work · label units

1. Evaluate each expression.

a. $6 + 4 \times 5 =$ ____

b. $(6 + 4) \times 5 =$ ____

c. $3^2 + 2 \times 4 =$ ____

d. $48 \div (2 + 6) =$ ____

e. $5^2 - 10 \div 2 =$ ____

f. $36 \div 3 \times 2 =$ ____

2. Add one pair of parentheses to make each statement true.

a. $2 + 3 \times 4 = 20$

b. $24 \div 4 + 2 = 4$

3. **Tile Floor.** A square floor is **8 tiles by 8 tiles**. Mr. Ruiz replaces all but **4** tiles. Each new tile costs **$3**. Evaluate $3 \times (8^2 - 4)$ to find the cost.

Step 1 Inside ( ): $8^2 = 64$, then subtract 4.

Step 2 Multiply that by 3.

Show your work.

Answer_____ dollars

4. Bead Kits. Ms. Lewis has 5 bags with 4^2 beads in each bag. Her class uses 30 beads. Evaluate $5 \times 4^2 - 30$ to find how many beads are left.

Step 1 Find 4^2 first.

Step 2 Multiply by 5, then subtract 30.

Answer_____ beads

5. Who Is Right? For $18 - 6 + 2$, Kim got 10 and Lou got 14. Who is correct?

Frame ___ *is correct because + and − are done* ___.

☐ Kim ☐ Lou

TEST PRACTICE

choose the best answer

6. What is the value of $5 + 3 \times 2^2$?

A 17 **B** 32 **C** 41 **D** 256

7. Which expression has a value of 14?

A $(2 + 3) \times 4$ **B** $2 + 3 \times 4$ **C** $2^3 + 4$ **D** $16 \div 2 \times 4$

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Show someone at home why $2 + 3 \times 4$ and $(2 + 3) \times 4$ have different values.